

Money Plot Labs Technical Whitepaper

The Pure Math of Financial Freedom

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Abstract

This whitepaper derives the explicit, closed-form analytical solution for a fixed-horizon retirement timeline. By presenting a simple linear model first, we establish a transparent baseline using middle-school algebra. We then expand this model into exponential growth curves by mapping the exact continuous boundaries between an asset accumulation phase and an annuity-due depletion phase. This isolates the exact number of required working years (w) as a function of savings rate, spending targets, initial capital, investment growth, and desired legacy floors.

1 Introduction & Variables

Standard personal finance paradigms rely on fragile rules of thumb. To achieve absolute mathematical certainty, we model an individual's financial lifecycle across a strictly bounded timeline from a starting age (T_{start}) to a maximum life expectancy (T_{end}). We define the total planning horizon as:

$$H = T_{\text{end}} - T_{\text{start}} \quad (1)$$

Within this horizon H , an individual works for w consecutive years and spends the remaining f years in full retirement, such that:

$$H = w + f \quad (2)$$

1.1 Parameter Definitions

- A_0 : Initial Net Worth / Starting Balance (\$)
- s : Annual Savings during working years (\$)
- c : Annual Consumption during retirement years(\$)
- L : Desired Capital Legacy Floor / Inheritance Remaining at Death (\$)
- r : Real Growth Rate adjusted for inflation
- w : Accumulation Phase duration (Working Years)
- f : Depletion Phase duration (Retirement Years)

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2 The Linear Baseline Case ($r = 0\%$)

Before introducing the compounding curves of financial markets, we establish intuitive baselines by assuming real asset growth perfectly tracks inflation ($r = 0\%$). In this environment, wealth accumulates and depletes along perfectly straight lines, making it easy to isolate how system parameters structurally shift timeline horizons.

2.1 Case 1: The Idealized Zero-Baseline Framework ($A_0 = 0, L = 0$)

We begin with a simplified world containing a zero starting balance ($A_0 = 0$) and no terminal inheritance target ($L = 0$).

During the working phase, assets grow linearly via cash savings flow c injected at the end of each period for w years:

$$\text{Peak Assets Case 1} = s \cdot w \quad (3)$$

Upon entering retirement, accumulation stops depletion begins draining our peak nest egg by a flat annual budget block b for the remaining f retirement years:

$$\text{Peak Assets Required Case 1} = c \cdot f \quad (4)$$

To locate the precise intersection boundary where working ends and retirement begins, we enforce our system constraint where the peak accumulated capital must exactly equal the total retirement drawdown:

$$s \cdot w = c \cdot f \quad (5)$$

Substituting our timeline identity $f = H - w$ allows us to solve for w :

$$s \cdot w = c \cdot (H - w) \quad (6)$$

Grouping terms containing our target variable w onto the left-hand side and factoring yields:

$$w(s + c) = c \cdot H \quad (7)$$

Dividing through isolates our fundamental idealized baseline relationship:

$$w = \frac{c \cdot H}{s + c} \quad (8)$$

In the example in Figure 1, a person saving 25% of their income: $c = \$45,000/\text{year}$, $s = \$15,000/\text{year}$, $T_{start} = 25$ years, $T_{end} = 95$ years.

$$\begin{aligned} w &= \frac{\$45,000 \cdot 70 \text{ years}}{\$45,000 + \$15,000} \\ w &= 52.5 \text{ years} \end{aligned}$$

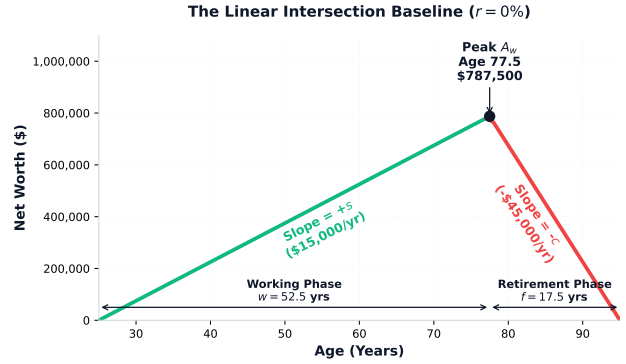


Figure 1: The straight-line boundary crossover mapping where $r = 0\%$. Assets accumulate smoothly via **savings** and deplete symmetrically via **consumption** until hitting the legacy boundary.

2.2 Case 2: The Parameterized Linear Solution with Boundary Modifiers ($A_0 \neq 0, L \neq 0$)

We now expand our straight-line environment to account for real-world initial endowments and estate conservation goals. We reintroduce the asset head-start parameter (A_0) and the designated capital legacy floor (L).

Incorporating these conditions alters our primary boundary accumulation and depletion states:

$$\text{Peak Assets Case 2} = A_0 + s \cdot w \quad (9)$$

$$\text{Peak Assets Required Case 2} = L + c \cdot f \quad (10)$$

Setting our parameterized accumulation trajectory equal to the absolute terminal requirement forces the system boundary equation:

$$A_0 + s \cdot w = L + c \cdot f \quad (11)$$

Substituting our timeline identity $f = H - w$ and isolating our targeted timeline variables onto the left-hand side of the identity yields:

$$s \cdot w + c \cdot w = c \cdot H + L - A_0 \quad (12)$$

Factoring out w collects our parameters into clean operational dimensions:

$$w(s + c) = c \cdot H + L - A_0 \quad (13)$$

Dividing through by our total structural velocity ($s + c$) delivers the complete **Parameterized Linear Equation**:

$$w = \frac{c \cdot H + L - A_0}{s + c} \quad (14)$$

This intermediate step clearly demonstrates the pure algebraic translation of our boundary mechanics: an initial balance (A_0) directly subtracts from required working years, while a desired legacy floor (L) extends the required labor timeline.

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3 Phase 1: The Exponential Accumulation Engine

When we introduce real market investment returns ($r > 0\%$), our straight lines transform into curves. During the working phase (w), net worth grows via compound interest on the initial principal while absorbing an ordinary annuity flow of annual savings c injected at the end of each period. The net worth at the exact boundary of retirement (A_w) is defined by:

$$A_w = A_0(1 + r)^w + s \cdot \frac{(1 + r)^w - 1}{r} \quad (15)$$

3.1 Step-by-Step Inductive Derivation of A_w

To understand the structural origin of the accumulation equation, we trace the boundary state changes sequentially at the close of each annual compounding cycle:

$$\begin{aligned}
A_1 &= A_0(1+r) + s \\
A_2 &= A_1(1+r) + s = [A_0(1+r) + s](1+r) + s \\
&= A_0(1+r)^2 + s(1+r) + s \\
A_3 &= A_2(1+r) + s = [A_0(1+r)^2 + s(1+r) + s](1+r) + s \\
&= A_0(1+r)^3 + s(1+r)^2 + s(1+r) + s
\end{aligned}$$

Extending this inductive sequence to a generalized working timeline of w years yields:

$$A_w = A_0(1+r)^w + s \sum_{k=0}^{w-1} (1+r)^k \quad (16)$$

The summation block represents a classic geometric progression. Applying the standard Geometric Series Summation Theorem where the common ratio is $(1+r)$, the series condenses analytically:

$$\sum_{k=0}^{w-1} (1+r)^k = \frac{(1+r)^w - 1}{(1+r) - 1} = \frac{(1+r)^w - 1}{r} \quad (17)$$

Substituting Equation 18 back into Equation 17 yields our verified final accumulation master expression matching Equation 16.

4 Phase 2: The Exponential Depletion Engine

Upon entering retirement, the cash flow behavior flips. To match real-world physical constraints, living expenses must be accessible at the *beginning* of each calendar year. Therefore, retirement spending is modeled strictly as an **Annuity Due**.

$$R_f = A_w(1+r)^f - c \cdot \frac{(1+r)^{f+1} - (1+r)}{r} \quad (18)$$

4.1 Step-by-Step Inductive Derivation of the Depletion Phase

We trace the remaining capital balance R at the close of each retirement year sequentially, demonstrating why the timing mismatch injects an extra factor of $(1+r)$ compared to an ordinary annuity:

$$\begin{aligned}
R_1 &= (A_w - c)(1+r) = A_w(1+r) - c(1+r) \\
R_2 &= (R_1 - c)(1+r) = [A_w(1+r) - c(1+r) - c](1+r) \\
&= A_w(1+r)^2 - c(1+r)^2 - c(1+r) \\
R_3 &= (R_2 - c)(1+r) = [A_w(1+r)^2 - c(1+r)^2 - c(1+r) - c](1+r) \\
&= A_w(1+r)^3 - c(1+r)^3 - c(1+r)^2 - c(1+r)
\end{aligned}$$

Extending this progression to a total retirement duration of f years yields:

$$R_f = A_w(1+r)^f - c \sum_{k=1}^f (1+r)^k \quad (19)$$

Factoring out a common term of $(1+r)$ isolates a standard geometric progression inside the summation block:

$$\sum_{k=1}^f (1+r)^k = (1+r) \sum_{k=0}^{f-1} (1+r)^k = (1+r) \cdot \frac{(1+r)^f - 1}{r} \quad (20)$$

Distributing the factored $(1+r)$ back across the numerator yields the completed, time-synchronized depletion engine expression matching Equation 19.

5 The Unified Derivation with Legacy Capital Floors

To complete our global solution, we enforce our final boundary constraint where our asset base at the end of life must exactly match our legacy target ($R_f = L$). By substituting our timeline identity $f = H - w$, we can map our accumulation peak directly into our depletion model.

5.1 The Color-Coded Substitution

We inject the accumulation boundary definition (Equation 16) directly into our synchronized depletion phase expression (Equation 19), creating a unified timeline equation:

$$\left[A_0(1+r)^w + s \cdot \frac{(1+r)^w - 1}{r} \right] (1+r)^{H-w} - c \cdot \frac{(1+r)^{H-w+1} - (1+r)}{r} = L \quad (21)$$

5.2 The Algebraic Isolation

Distributing the central exponential term $(1+r)^{H-w}$ across our accumulation bracket allows the localized w variables to elegantly combine and drop out of the leading terms:

$$A_0(1+r)^H + \frac{s}{r}(1+r)^H - \frac{s}{r}(1+r)^{H-w} - c \cdot \frac{(1+r)^{H-w+1} - (1+r)}{r} = L \quad (22)$$

Multiplying the entire equation by our common denominator r removes fractional complexity:

$$rA_0(1+r)^H + s(1+r)^H - s(1+r)^{H-w} - c(1+r)^{H-w+1} + c(1+r) = rL \quad (23)$$

Grouping the terms containing our targeted timeline variable w onto the right-hand side, and moving our legacy parameter to the left-hand side:

$$(1+r)^H (rA_0 + s) + c(1+r) - rL = [s + c(1+r)] (1+r)^{H-w} \quad (24)$$

Isolating our exponential timeline block completely:

$$(1+r)^{H-w} = \frac{(rA_0 + s)(1+r)^H + c(1+r) - rL}{s + c(1+r)} \quad (25)$$

Taking the natural logarithm ($\ln$) of both sides frees our timeline variable from the exponent:

$$(H - w) \ln(1 + r) = \ln \left[\frac{(rA_0 + s)(1 + r)^H + c(1 + r) - rL}{s + c(1 + r)} \right] \quad (26)$$

Isolating w yields the final closed-form analytical solution for the required working timeline:

$$w = H - \frac{\ln \left[\frac{(rA_0 + s)(1 + r)^H + c(1 + r) - rL}{s + c(1 + r)} \right]}{\ln(1 + r)} \quad (27)$$

Observe that evaluating the limit of Equation 28 as $r \rightarrow 0$ collapses perfectly into our baseline parameterized linear case derived in Equation 15, confirming absolute mathematical symmetry across all domains.

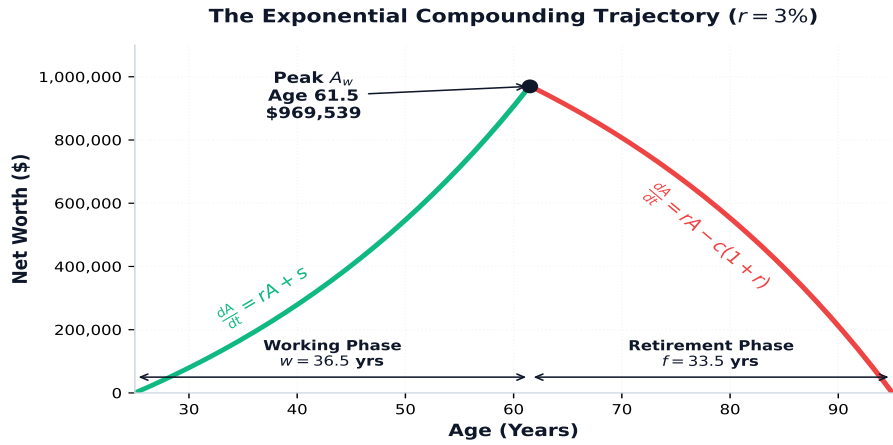


Figure 2: The exponential compounding lifecycle curves where $r > 0\%$. The **accumulation** phase curves upward via geometric expansion and transitions to an accelerated annuity-due **depletion** phase.

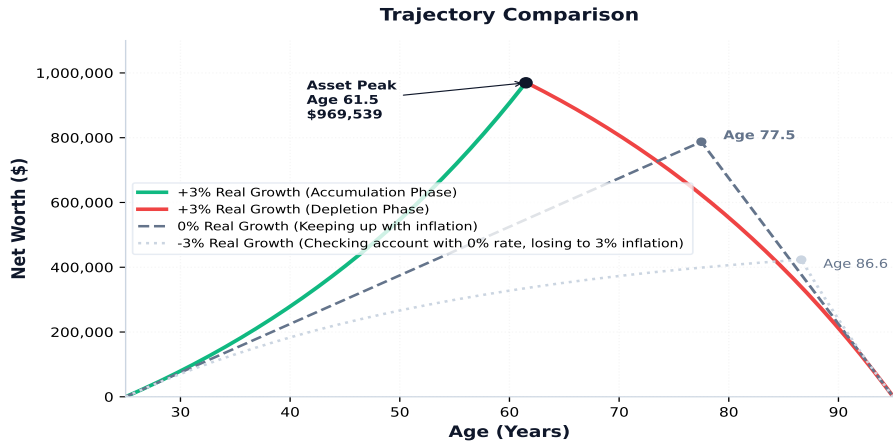


Figure 3: The exponential compounding curve compared to both assets just keeping with inflation and assets losing to inflation.